

# Introduction

The KiCad Team

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## Feedback

The KiCad project welcomes feedback, bug reports, and suggestions related to the software or its documentation. For more information on how to submit feedback or report an issue, please see the instructions at <https://www.kicad.org/help/report-an-issue/>

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# Welcome

KiCad is a free and open-source electronics design automation (EDA) suite. It features schematic capture, integrated circuit simulation, printed circuit board (PCB) layout, 3D rendering, and plotting/data export to numerous formats. KiCad also includes a high-quality component library featuring thousands of symbols, footprints, and 3D models. KiCad has minimal system requirements and runs on Linux, Windows, and macOS.

KiCad 8.0 is the most recent major release. It includes hundreds of new features and bug fixes. Some of the most notable new features include:

- A Properties Panel in the Schematic, Symbol, and Footprint editors, which lets you quickly view and edit properties of selected objects. This works even when multiple different types of objects are selected at once. This expands on the Properties Panel in the PCB editor, which was introduced in version 7.0.
- HTTP libraries, which allow connecting KiCad to a networked component database or ERP system via a REST API. The REST endpoint can provide symbol and footprint mappings and other metadata.
- A search panel in the Schematic editor, which lets you search for symbols, labels, and schematic text. This mirrors the search panel in the PCB editor, which was added in version 7.0.
- The Symbol Fields Table has been expanded and now includes a fully-featured and customizable BOM export tool. The legacy BOM export tool, which uses external scripts to generate a BOM in the desired format, is still available.
- The SPICE simulator interface has been expanded and overhauled. The simulator tool now supports several new analysis types (FFTs, noise simulations, pole-zero simulations, and s-parameter analyses). The simulator result plots can now include user-defined signals, including mathematical operations, and the cursor and waveform measurement tools have been improved. Operating point analysis results can be displayed on the schematic itself to show the simulated operating point of each node.
- The command-line interface has been expanded, and now includes support for running ERC/DRC, exporting to more formats, and specifying text variables, among other improvements.
- A net navigator tool has been added in the Schematic editor, which displays the locations of a highlighted net across a schematic hierarchy.
- Grids can now be automatically overridden for specific object types, for example to use a finer grid for moving text and a coarser grid for symbols.
- Power symbols are now easier to work with, as their assigned net is now determined by the symbol value rather than the pin name. Where before changing a power symbol's net required editing the symbol in the symbol editor, in version 8 you can change the net directly in the schematic.
- A graphical comparison tool has been added to compare symbols and footprints between libraries and the copies embedded in schematics and PCBs.
- Board designs can be exported to the IPC-2581 format, which includes complete fabrication and assembly information in a single file.
- The interactive length tuning tools in the PCB editor have been overhauled. Length tuning meanders are now persistent objects that can be easily modified after they are initially created.

Symbol fields are now synced to footprints in the PCB, so schematic symbol metadata is available in the board design.

- Graphical polygons on copper PCB layers can now have nets assigned, which can be useful for creating electrical structures with a precise shape.
- "Pin helper" tools have been added to the Schematic editor to quickly break out symbol pins to wires, nets, or no-connects.
- The 3D Viewer now has an appearance panel that improves user control over how board designs are visualized in 3D.
- New importers and improvements to existing importers. Version 8 adds support for importing EasyEDA (JLCEDA) Standard and Professional Edition projects and libraries, CADSTAR symbol and footprint libraries, Solidworks PCB board files, Altium Designer symbol and integrated libraries, EAGLE symbol libraries, and LTSpice schematics.
- New and improved exporters, including copper geometry in STEP exports and Cadence Allegro netlist export.
- Significant improvements to the Symbol, Footprint, and 3D Model libraries.

A full listing of new features and changes in KiCad 8.0 can be found [here](#).

# Installing and Upgrading KiCad

KiCad maintains compatibility and support with the maintained versions of Microsoft Windows, Apple macOS, and a number of Linux distributions. Some platforms have specific installation or upgrade instructions. Always check <https://www.kicad.org/download/> for the latest release information and instructions for your platform.

KiCad may compile and run on platforms that are not officially supported. The KiCad development team makes no guarantees that KiCad will continue to work on these platforms in the future. See <https://www.kicad.org/help/system-requirements/> for more details on supported platforms and hardware requirements.

KiCad uses a "major.minor.point" release version format. Major releases bring new features and other significant changes to the code. Minor releases are relatively rare and typically bring bug fixes that are too complicated for a point release. Point releases contain only bugfixes. Users are encouraged to update to the latest point release for their current major.minor version promptly, as these releases will not break file compatibility. Major releases almost always come with changes to file formats. KiCad is in general always backwards compatible with files created by older versions, but not forwards compatible: Once files are edited and saved by a new major version, these files will not be openable by the previous major version.

## Migrating from Previous Versions

In general, to migrate a design to a new version of KiCad, simply open the project with the new version, then open the schematic and PCB and save each file. More details about specific issues that may come up when migrating designs is covered in the Schematic Editor and PCB Editor chapters of the manual.

### NOTE

Make sure to save a backup of your design before opening it with a new version of KiCad. Once saved in a new major version of KiCad, designs can no longer be opened by previous major versions.

The symbol library format changed in KiCad 6.0. To continue editing symbol libraries made with a previous version of KiCad, these libraries need to be migrated to the new format. For details on this process, see the Schematic Editor chapter of the manual. Symbol libraries that have not been migrated can still be opened and used in read-only mode.

# KiCad Workflow

This section presents a high-level overview of the typical KiCad workflow. Note that KiCad is a flexible software system, and there are other ways of working that are not described here. For more information about each of the steps described in this section, please see the later chapters in this manual.

## NOTE

A number of tutorials and guided lessons in using KiCad have been created by community members. These resources can be a good way to learn KiCad for some new users. See the Further Reading section at the end of this chapter for more information.

## Basic Terminology

KiCad uses a number of terms that are fairly standard in the area of electronics design automation (EDA) software, and some that are more specific to KiCad. This section lists some of the most common terms used throughout KiCad's documentation and user interface. Other terms that are more specific to a certain part of the KiCad workflow are defined later in this manual.

A **schematic** is a collection of one or more pages (sheets) of circuit schematic drawings. Each KiCad schematic file represents a single sheet.

A **hierarchical schematic** is a schematic consisting of multiple pages nested inside each other. KiCad supports hierarchical schematics, but there must be a single **root sheet** at the top of the hierarchy. Sheets within a hierarchy (other than the root sheet) may be used more than once, for example to create repeated copies of a subcircuit.

A **symbol** is a circuit element that can be placed on a schematic. Symbols can represent physical electrical components, such as a resistor or microcontroller, or non-physical concepts such as a power or ground rail. Symbols have **pins** which serve as the connection points that can be wired to each other in a schematic. For physical components, each pin corresponds to a distinct physical connection on the component (for example, a resistor symbol will have two pins, one for each terminal of the resistor). Symbols are stored in **symbol libraries** so they can be used in many schematics.

A **netlist** is a representation of a schematic that is used to convey information to another program. There are many netlist formats used by various EDA programs, and KiCad has its own netlist format that is used internally to pass information back and forth between the schematic and PCB editors. The netlist contains (among other things) all the information about which pins connect to each other, and what name should be given to each **net**, or set of connected pins. Netlists can be written to a **netlist file**, but in modern versions of KiCad, this is not necessary as part of the normal workflow.

A **printed circuit board**, or PCB, is a design document that represents the physical implementation of a schematic (or technically, a netlist). Each KiCad board file refers to a single PCB design. There is no official support for creating arrays or panels of PCBs within KiCad, although some community-created add-ons provide this functionality.

A **footprint** is a circuit element that can be placed on a PCB. Footprints often represent physical electrical components, but can also be used as a library of design elements (silkscreen logos, copper antennas and coils, etc.). Footprints can have **pads** which represent copper areas that are electrically-connected. The netlist will associate symbol pins with footprint pads.

A **worksheet** is a drawing template, typically containing a title block and frame, that is used as the template for schematic sheets and PCB drawings.

**Plotting** is the process of creating manufacturing outputs from a design. These outputs may include machine-readable formats such as Gerber files or pick-and-place listings, as well as human-readable formats such as PDF drawings.

**Ngspice** is a mixed-signal circuit simulator, originally based on Berkeley SPICE, that is integrated into KiCad's schematic editor. By using symbols with attached SPICE models, you can run circuit simulations on KiCad schematics and plot the results graphically.

## KiCad Components

KiCad consists of a number of different software components, some of which are integrated together to facilitate the PCB design workflow, and some of which are standalone. In early versions of KiCad, there was very little integration between the software components. For example, the schematic editor (historically called Eeschema) and PCB editor (historically called PcbNew) were separate applications that had no direct link, and to create a PCB based on a schematic, users had to generate a netlist file in Eeschema and then read this netlist file in PcbNew. In modern versions of KiCad, the schematic and PCB editor are integrated into the KiCad project manager, and using netlist files is no longer required. Many tutorials still exist that refer to the old KiCad workflow of separate applications and netlist files, so be sure to check the version being used when looking at tutorials and other documentation.

The main KiCad components are usually started from the launcher buttons in the KiCad project manager window. These components include:

| Component name     | Description                                                                     |
|--------------------|---------------------------------------------------------------------------------|
| Schematic Editor   | Create and edit schematics; simulate circuits with SPICE; generate BOM files    |
| Symbol Editor      | Create and edit schematic symbols and manage symbol libraries                   |
| PCB Editor         | Create and edit PCBs; export 2D and 3D files; generate fabrication output files |
| Footprint Editor   | Create and edit PCB component footprints and manage footprint libraries         |
| GerbView           | Gerber and drill file viewer                                                    |
| Bitmap2Component   | Convert bitmap images to symbols or footprints                                  |
| PCB Calculator     | Calculator for components, track width, electrical spacing, color codes, etc.   |
| Page Layout Editor | Create and edit worksheet files                                                 |

## User Interface

KiCad has a number of user interface behaviors that are common to all the different editor windows. Some of these behaviors are described in more detail in later chapters of this manual.

Objects can be selected by clicking on them or by dragging a selection window around them. Dragging from left to right will result in a selection of any items that are completely within the window. Dragging from right to left will result in a selection of any items that touch the window. Pressing certain modifier keys while



clicking or dragging will change the selection behavior. These keys are platform-specific and are described in the Editing Options section of the Preferences dialog.

KiCad editors have the concept of a **tool** which can be thought of as a mode that the editor is in. The default tool is the selection tool, which means that clicking will select objects under the mouse cursor. There are also tools for placing new objects, inspecting existing objects, etc. The active tool is highlighted in the toolbar, and the name of the active tool is shown in the bottom right of the editor in the status bar. Pressing  always means "cancel" in KiCad: if a tool is in the middle of some action (for example, routing tracks), the first press of  will cancel that action. The next press of  will exit the tool completely, returning to the default selection tool. With the selection tool active, pressing  will clear the current selection, if one exists.

## KiCad Projects and Files

### NOTE

This section of the KiCad documentation has not yet been written. We appreciate your patience as our small team of volunteer documentation writers work to update and expand the documentation.

- File types and project structure
- Project workflow
- Schematic <> PCB workflow
- Standalone vs. project mode for schematic and PCB editors

## Symbol and Footprint Libraries

### NOTE

This section of the KiCad documentation has not yet been written. We appreciate your patience as our small team of volunteer documentation writers work to update and expand the documentation.

- Relationship between libraries and design files
- Global vs project libraries
- The KiCad library project (built-in global libraries)

## Accessory Tools

### NOTE

This section of the KiCad documentation has not yet been written. We appreciate your patience as our small team of volunteer documentation writers work to update and expand the documentation.

- GerbView
- PCB Calculator
- Bitmap2Component
- Worksheet Editor (pl\_editor)

# Further Reading

The latest version of this manual can be found in multiple languages at <https://docs.kicad.org> Manuals for previous versions of KiCad can also be found at that website.

The KiCad user community includes a number of forums and chat platforms that are operated independently from the KiCad development team but are fully endorsed as a great way to find help with problems, learn tips and tricks, and share examples of KiCad projects. A listing of community resources is available under the Community heading at <https://www.kicad.org>

Users interested in compiling KiCad from source and/or contributing to KiCad development should visit our developer documentation site at <https://dev-docs.kicad.org> for instructions, policies and guidelines, and technical information about the KiCad codebase.